

SHRIDHAR PATHAK

Senior Director of AI — Multi-Agent Systems & Enterprise GenAI

Cambridge, MA | shripathmit@gmail.com | +1-214-794-6073 | linkedin.com/in/shridhar-pathak-19277927 | github.com/shripathmit | shridharpathak.me

PROFESSIONAL SUMMARY

Senior Director of AI with 14+ years building large-scale AI platforms and multi-agent conversational systems. Currently leading enterprise GenAI strategy and engineering at Moderna across 20+ applications; previously architected Alexa's first multi-agent conversational framework serving millions of users globally. Track record of scaling high-reliability, low-latency AI infrastructure and leading global organizations (35+ engineers across 8+ countries) from 0-to-1 builds to enterprise rollout in regulated industries.

SKILLS

LLM Orchestration, Multi-Agent Systems, Conversational AI, Retrieval-Augmented Generation (RAG), Fine-Tuning & Distillation, Function Calling, MLOps, Speech Recognition, NLP, AI Governance & Enterprise Compliance, AWS, Distributed Microservices, Python, Low-Latency Serving, Engineering Org Design, Global Delivery, C-Suite Stakeholder Management

PROFESSIONAL EXPERIENCE

Senior Director, AI & ML — Moderna, Cambridge, MA

Jul 2023 – Present

- Defined and executed Moderna's generative AI roadmap across 20+ enterprise applications (R&D, manufacturing, clinical, quality), spanning conversational agents, function calling, and LLM orchestration workflows.
- Built and led a 35+ engineer cross-functional AI/ML organization delivering 30+ production LLM applications on AWS with Python and MLOps pipelines.
- Architected multi-agent conversational systems leveraging fine-tuning, distillation, and RAG to automate knowledge work across regulated document workflows.
- Partnered with C-suite and business unit leaders to translate AI capabilities into measurable business outcomes; established strategic partnerships with OpenAI, AWS, and cloud AI vendors.

Senior Manager & Tech Lead, Alexa — Amazon, Seattle, WA

Jun 2022 – Jul 2023

- Engineering leader for large-scale conversational AI and multi-agent orchestration frameworks powering Alexa globally: 35+ direct reports and 110+ partner-team reports across 8+ countries.
- Architected and launched Alexa's first multi-agent conversational switching framework, orchestrating celebrity personalities (Samuel L. Jackson, Melissa McCarthy, Shaquille O'Neal, Amitabh Bachchan) and partner agents (Disney).
- Designed contextual aggregation and routing services handling millions of low-latency voice requests daily; set architectural standards for reliability and developer extensibility.
- Drove cross-org alignment across Alexa AI, retail, device, and international engineering teams.

Principal Technical Program Manager & Tech Lead, Alexa — Amazon, Seattle, WA

Jan 2021 – Jun 2022

- Pioneered the first multi-agent conversational AI framework, enabling monetization and engagement measurement for celebrity personalities.
- Launched Alexa Voice Shopping across 10 countries (US, India, Japan, Canada, France, Italy, Spain, Mexico, Ireland, Australia), covering localization, regulatory adaptation, and infrastructure scaling.
- Led integration of speech recognition, NLP, contextual understanding, and response orchestration; defined APIs and service endpoints enabling ML-powered interactions in the Amazon India retail app (millions of DAU).

PROFESSIONAL EXPERIENCE — CONTINUED

Senior Technical Program Manager, Alexa India — Amazon, Bengaluru, India Jan 2019 – Dec 2020

- Founding technical member of the Alexa India Retail team; built the organization from scratch to 16+ engineers and analysts.
- Led integration of Alexa into the Amazon India retail app, architecting the APIs and ML endpoints that enabled millions of users to shop via voice.
- Established team culture and best practices for retail voice analytics in a new market.

Manager & Tech Lead, Amazon Retail (Southeast Asia Launch — “Project Nautilus”) — Amazon, Seattle, WA Jun 2016 – Dec 2018

- Single-threaded technical owner for Amazon's first retail entry into Southeast Asia, orchestrating roadmap alignment across 26 independent teams and 7 global organizations.
- Reduced the standard three-year country launch timeline to nine months by decoupling legacy US-centric systems into a lightweight, region-specific operating model.
- Architected the region's first end-to-end digital marketing and last-mile logistics stack, including proprietary two-hour delivery in Singapore fulfilling 100% of regional orders via proprietary systems.

Software Development Engineer (SDE-1 to SDE-2, Tech Lead) — Amazon Jun 2012 – Jun 2016

- Built distributed services for Amazon Register (POS platform): merchant onboarding, payments, and integration systems.
- Backend developer on Amazon Digital Video (encoding area); transaction processing and fraud control; data warehouse pipelines supporting analytics.
- Scrum Master for a two-pizza team, coordinating delivery across sites.

Junior Research Fellow — Tata Institute of Fundamental Research (TIFR), Mumbai, India Jul 2009 – Apr 2010

- GRAPES-3 cosmic-ray experiment (CERN framework): designed detector-monitoring algorithms in C++ and ROOT, improving monitoring accuracy by ~15%.
- Built a web interface for VME control, configuration, testing, and monitoring.

ADVISORY

Tech Advisor — Noesis (Enterprise Agent Workforce Platform) 2025 – Present

- Architectural blueprint for enterprise-scale AI agent orchestration: Policy-as-Code, runtime supervision, tool use, function calling, memory, and workflow chaining; performance benchmarking models for agent quality and enterprise ROI.

Tech & AI Advisor — EZbeat (by Health Galaxy) Sep 2023 – Present

EDUCATION

MIT Sloan School of Management — Executive MBA, Sep 2024 – May 2026

Harvard Business School — Cross-registration via MIT Sloan, Entrepreneurship, Jan 2026 – May 2026

University of Texas at Dallas — M.S., Computer Science, 2012

Indian Institute of Technology Bombay — Three Research Internships: Computer Science (Computer Vision and NLP labs) and Aerospace Engineering, 2005–2009

Pune Institute of Computer Technology — B.E., Computer Science, 2009

AWARDS & RECOGNITION

Stanford Seed Consultant, Stanford Graduate School of Business — Strategic advisor to high-growth founders

Engineering Leadership Development, Rice University

KVPY Young Scientist Fellowship, Government of India

Patent: Neo crankshaft for easy cycling